MATTEO CAVO

Rome, Italy | matteo.cavo.data@gmail.com | +39 351 714 62 39 | LinkedIn | GitHub | Portfolio

Data and Business Intelligence Analyst

Data Engineering · Applied ML · AI & GenAI Workflows

PROFESSIONAL PROFILE

Data & Business Intelligence Analyst with hands-on experience in ETL pipeline design, data workflow automation and end-to-end BI systems, including 5+ years in content operations and metadata quality at Gracenote (Nielsen), Amsterdam. Skilled in building data models, Power BI dashboards and analytical workflows using Python, SQL and DuckDB, with hands-on experience in Databricks, PySpark, Delta Lake and applied machine learning. Capable of translating complex data into actionable insights and supporting data-driven decision-making across international teams. Additional hands-on use of GenAI tools including ChatGPT, Claude, Codex, Gemini, Google AI Studio, GitHub Copilot, Databricks Genie and Antigravity for AI-assisted development, workflow automation and analytical decision support.

AREAS OF EXPERTISE

Business Intelligence | Data Engineering & Automation | Data Quality & Governance | Advanced Analytics | Applied Machine Learning | AI & GenAI Workflows | Exploratory Data Analysis (EDA) | Operational Analytics | Data Storytelling & Insight Communication | Data-driven Decision Making | Analytical Thinking | Communication & Presentation Skills | Feature Engineering

TECHNICAL PROFICIENCIES

- **Data Analytics & BI:** SQL (JOIN, GROUP BY, CTE, Subqueries, Window Functions) | Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn) | Jupyter Notebook (EDA, ML Experimentation, Documentation) | Microsoft Power BI (DAX, Power Query, Data Modelling, Power BI Service) | Microsoft Excel (Advanced formulas, Pivot Tables, Power Query) | Git/GitHub
- **Data Engineering & Automation:** Databricks · PySpark · Delta Lake (Bronze-Silver-Gold) · MLflow | ETL/ELT pipeline design (API, open data, versioned datasets) | Data lifecycle management (RAW - CLEAN - MART) | Feature engineering | DuckDB for analytical workloads | YAML-driven pipeline configuration | Star schema & KPI layer modelling
- **Machine Learning & Advanced Analytics:** Supervised Learning (Regression, Classification, Decision Trees, Random Forests, Gradient Boosting, XGBoost) | Unsupervised Learning (K-means) | Feature Engineering | Statistical Inference & Hypothesis Testing | Cross-Validation | Model Evaluation (Precision, Recall, F1, ROC-AUC)
- **AI & GenAI Workflows:** Structured prompting · Prompt refinement · LLM-assisted analytics · AI-assisted development · Structured outputs · Workflow orchestration · Decision-support automation | ChatGPT · Claude · Codex · Gemini · Google AI Studio · GitHub Copilot · Databricks Genie · Antigravity
- **Web & CMS Technologies:** HTML, CSS, PHP | CMS platforms (WordPress & similar) | Google Analytics (GA4)
- **Creative & Design Tools:** Adobe Photoshop | Adobe Illustrator

PROFESSIONAL EXPERIENCE

Data & Business Intelligence Analyst **2025 - Present**

Independent Data & Analytics Projects

- Designed and developed ETL/ELT pipelines from APIs and open data (Python, SQL)
- Built end-to-end Databricks pipeline with PySpark, Delta Lake (Bronze-Silver-Gold), feature engineering and MLflow
- Applied ML: trained, evaluated and interpreted predictive models (Scikit-learn, XGBoost)
- Automated data workflows using RAW - CLEAN - MART architecture
- Built data models and Power BI dashboards (DAX, KPI layer)
- Used DuckDB for analytics workloads on complex datasets; developed reproducible workflows with Jupyter Notebook and GitHub
- AI-assisted development and workflow automation using ChatGPT, Claude, Codex

Content Operations Co-ordinator / Editor SVC/VD Europe **2021 - August 2026**

Gracenote (Nielsen) | Amsterdam, Netherlands

As Editor SVC/VD Europe: ensured accurate descriptors and video metadata for on-demand catalogues by assessing, designing, and proposing editorial, procedural, and operational solutions. Oversaw planning, commissioning, editing, and publication of written and multimedia content while maintaining editorial standards, quality, and regional relevance. Collaborated with writers, designers, producers, editors, and international teams to improve workflows, metadata consistency, and operational quality.

As Content Operations Co-ordinator: optimised content discoverability and personalised recommendations by curating descriptors for high-profile international movies and series. Carried out detailed research to enrich media content, artist, and production information for enhanced database completeness. Classified content with precise genres, themes, and relevant tags to augment searchability and recommendation accuracy. Enforced consistent formatting and style guidelines, maintaining database integrity and usability.

Key Achievements:

- Increased internal data quality KPIs by 15% through systematic classification, metadata standardisation, and workflow improvement
- Improved metadata accuracy and database reliability across 500K+ multimedia titles for global platforms including Netflix, Amazon Prime, Disney+, Sky, Mediaset Infinity, and RAI
- Reduced processing time and improved cross-border workflow efficiency by streamlining operations across international teams
- Enhanced data-driven content strategy and catalogue accuracy by evaluating performance metrics and strengthening metadata quality KPIs across large-scale entertainment libraries
- Improved operational quality by monitoring metadata accuracy, identifying recurring issues, and supporting corrective actions across large-scale video catalogues
- Accelerated onboarding and knowledge transfer through structured training and mentoring of junior editors on Video Descriptors, metadata standards, internal tools, and editorial workflows
- Served as a reference point for Video Descriptors and editorial workflows across European teams, supporting daily problem-solving, best-practice sharing, and cross-team collaboration

Music Producer · Label Founder · CEO & Creative Director **2005 - 2024**

Maztek / Subculture Records / 0101 Music

Delivered high-quality music production and visual design from concept to final release. Enhanced brand visibility and engagement by leading international digital projects and branding campaigns. Founded Subculture Records (2005) and 0101 Music (2018), independent labels with international distribution and collaborations. Managed independent labels, co-ordinated international artists and partners, and defined long-term strategic goals.

- Organised 10K+ creative sessions for digital content development including music, graphics, web and video
- Elevated audience engagement by managing five social media pages and creating hundreds of innovative posts
- Raised operational performance by co-ordinating international artist and partner relations and scrutinising data to update key processes

Web & Multimedia Designer **2004 - 2010**

Sinopia / Other Companies | Rome, Italy

Produced multimedia and visual assets emphasising UX and digital engagement. Collaborated with marketing, development and content teams to align design with project goals.

- Enhanced user engagement by developing data-driven UX and visual layouts
- Built and deployed websites using HTML, CMS platforms and PHP to deliver functional and responsive web solutions
- Produced logos and graphic assets leveraging Adobe Creative Suite to strengthen brand identity

CERTIFICATIONS AND EDUCATION

Professional Certifications

- Google Advanced Data Analytics Professional Certificate (2026)
- Google AI Professional Certificate (2026)
- Databricks Fundamentals (2026)
- Microsoft Power BI Desktop for Business Intelligence - Udemy Business (2026)
- Microsoft Excel - Excel from Beginner to Advanced - Udemy Business (2025)

Certifications in Preparation

- Microsoft Power BI Data Analyst Associate (PL-300) - In Preparation
- Databricks Certified Data Engineer Associate - In Preparation

Education

- Multimedia IT Diploma - Accademia Pantheon Multimediale, Rome, 2003 (certificate of attendance and completion)
- High School Diploma - Istituto Marco Polo, Rome, Tourism track, 2005

DATA PROJECTS PORTFOLIO — GitHub | Portfolio

Rome Public Transport Reliability & Delay Prediction

End-to-end Databricks project analysing Rome's public transport reliability using GTFS Static and GTFS-Realtime data. Bronze-Silver-Gold architecture, PySpark/Delta Lake transformations, feature engineering, MLflow and machine learning models for delay analysis and service disruption risk estimation.

Databricks · PySpark · Delta Lake · MLflow · Python · SQL

Quaero v0.4.0 — Question-Driven Analytics Pipeline

Automated pipeline transforming raw datasets into analytical marts: profiling, cleaning, semantic inference, per-column data quality score, cleaning log, SQL export, Power BI export with DAX measures, column lineage with lineage HTML, dbt-style metrics. AI model disabled by default; AI interventions logged and traced.

Python · SQL · Power BI · DAX

Waste Collection & Urban Refuse Analysis (ISPRA)

Built structured data pipeline (RAW - CLEAN - MART) with YAML-driven configuration and DuckDB. Identification of "structurally virtuous" municipalities. Reproducible open-source workflow with GitHub versioning.

Python · Power BI · DAX

Streaming Competitive Benchmark

ETL pipeline from TMDB API; dimensional model with advanced KPIs: Genre Concentration Index, International Exposure, Jaccard Overlap. 5 platforms · 10+ markets.

Python · Power BI · DAX · NumPy

LANGUAGES

Italian, Native | English, Advanced (C1) | Spanish, Intermediate | Dutch & French, Basic

I authorise the processing of my personal data in accordance with Regulation (EU) 2016/679 (GDPR) and applicable Italian data protection legislation.